Learning Streaming Video Representation via Multitask Training

Learning Streaming Video Representation via Multitask Training

Yibin Yan*, Jilan Xu*, Shangzhe Di, Yikun Liu, Yudi Shi, Qirui Chen, Zeqian Li, Yifei Huang, Weidi Xie

School of Artificial Intelligence, Shanghai Jiao Tong University; Shanghai Innovation Institute; Fudan University; Shanghai AI Laboratory

* Equal Contribution • Project Page: https://go2heart.github.io/streamformer

OVERVIEW

StreamFormer: Streaming Video Backbone

VideoQA frames and answer

VideoQA

Question → Answer

- Learns representations at global, temporal, and spatial granularities
- Applicable to online action detection, online VIS, and VideoQA
- Trained via unified visual-language multitask alignment

Causal Temporal Attention | SigLIP Initialization | LoRA Spatial Adaptation

Supports diverse streaming video tasks effectively.

ABSTRACT

Key Contributions

- Backbone: Adds causal temporal attention to a pre-trained ViT (SigLIP) for efficient streaming.
- Training: Unifies diverse spatiotemporal tasks in a multitask visual-language framework.
- Results: Achieves competitive OAD, Online VIS, and VideoQA with low latency and high efficiency.

INTRODUCTION

Why Streaming Video Understanding?

- Real-time needs: robotics, autonomous driving, interactive systems
- Frame-by-frame processing without future frames
- Challenges:
  - Efficient per-frame computation
  - Long-term context preservation under memory/latency limits
  - Low-latency decision-making

MOTIVATION

Limitations of Framewise Image Encoders + LLM

- Coarse aggregation in LLMs loses fine spatiotemporal details
- Lack of early temporal modeling in vision backbone
- Need a backbone capturing global, temporal, and spatial cues

METHOD

Unified Multitask Visual-Language Alignment

- Three granularities: global (v), temporal (f), spatial (F)
- Tasks cast as video-text alignment with SigLIP text encoder
- Supports closed-set labels and open-ended captions

Alignment unifies tasks across granularities.

FORMULATION

Streaming Video Features and Alignment

- Input: V = [I1..IT]; computes v (global), f (per-frame), F (per-frame per-patch)
- Unifies N tasks Qi via alignment score Ai = g_tasks(visual, f_text(y))
- Benefits: single backbone, multitask transfer, extensibility

OUTPUTS

$v \in R^d$, $f \in R^{T\times d}$, $F \in R^{T\times H\times W\times d}$ (projected to shared space)

BACKBONE

Video Encoding with Causal Temporal Attention

- Patch embeddings + separate spatial/temporal position encodings
- Divided space-time attention blocks; temporal attention is causal
- Spatial attention initialized from SigLIP ViT; adapted with LoRA

TEMPORAL (CAUSAL)

- Accesses only past frames via causal mask
- Softmax along time; efficient for streaming

SPATIAL (PER-FRAME)

- Leverages strong SigLIP spatial priors
- LoRA enables lightweight adaptation

TEXT

Language Encoding with SigLIP

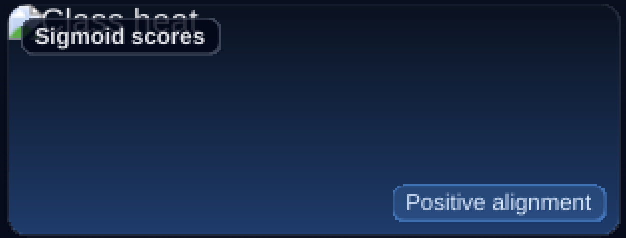
- Supports closed-set categories via prompts
- Supports open-ended captions/queries
- Text embeddings t = f_text(y) in shared space

Prompt: "a video clip of <action>"   Free-form description

TRAINING

Global-Level Objectives

- Action recognition (AR): uses last-frame feature v for causal aggregation
- Video-text retrieval (VTR): aligns v with captions
- Loss: sigmoid contrastive with learnable temperature and bias

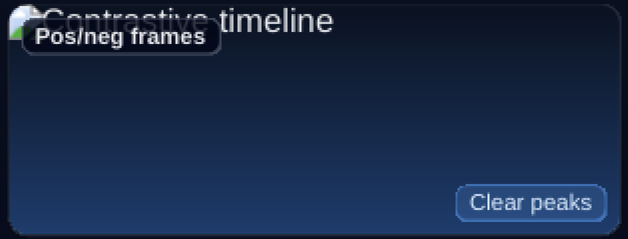


v captures clip-level semantics effectively.

TRAINING

Temporal-Level Objectives

- TAL: per-frame classification (action vs background)
- TVG: localizes segments from free-form queries
- Uses frame features f_t with contrastive formulation

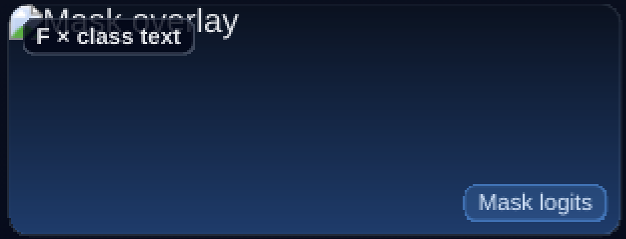


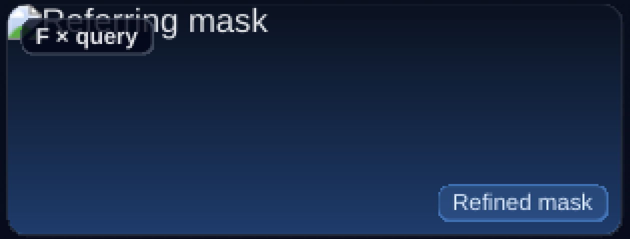
f_t encodes discriminative temporal cues.

TRAINING

Spatial-Level Objectives

- VOS (closed-set) and RVOS (referring language)
- Dot-product patch-text logits → upsample → cross-entropy
- Contrastive pixel-text alignment for RVOS





F preserves fine spatial details well.

OPTIMIZATION

Joint Objective and Training Strategy

- Total loss: L = L_global + L_temporal + L_spatial
- Alternates task sampling; accumulates gradients across tasks
- Balances tasks; avoids bias; single backbone learns all

SETUP

Pre-training Data and Implementation

DATASETS (~1M PAIRS)

- AR, VTR, TAL, TVG, VOS, RVOS
- Leverages rich human annotations

IMPLEMENTATION

- SigLIP base (patch16-224); LoRA rank 32
- Temporal layers random init + tanh gate
- 20 epochs, batch 128, lr 1e-6, 16 frames
- Backbone frozen for downstream heads

METRICS

Evaluation Metrics Used

- mAP (per-frame, THUMOS-14 OAD): mean of AP over classes; AP = area under precision–recall curve.
- AP (YTVIS19): average precision over IoU thresholds for instance masks; higher is better.
- AR1 (YTVIS19): average recall with 1 detection per video; higher is better.
- VideoMME (VideoQA): multi-skill video understanding benchmark; official: https://videomme.github.io/
- MLVU (VideoQA): long-form video understanding benchmark; official: https://mlvu-benchmark.github.io/

RESULTS

Online Action Detection (THUMOS-14)

Method	Backbone	w/ Flow	w/o Flow
MAT [2021]	SigLIP ViT	71.1	58.9
MAT [2021]	**StreamFormer**	**73.9**	**68.4**
LSTR [2021]	ResNet50	69.5	63.5
GateHUB [2023]	ResNet50	70.7	66.5

- Demonstrates state-of-the-art performance without flow; remains competitive with flow.

Baselines: MAT [96], LSTR [110], GateHUB [16].

RESULTS

Online Video Instance Segmentation (YTVIS19)

Method	Res.	Backbone	COCO-init.	AP	AR1
CTVIS [2024]	224p	ResNet50	No	43.6	42.7
CTVIS [2024]	224p	SigLIP ViT	No	40.9	41.8
CTVIS [2024]	224p	**StreamFormer**	No	**45.1**	**43.6**

- Surpasses ViT/ResNet baselines under identical settings; approaches high-res/COCO-initialized variants.

Baseline: CTVIS [116].

Table 2

RESULTS

Video Question Answering

Method (7B LMs)	Res.	VideoMME	MLVU
LLaVA-NeXT-Video [2024]	336^2	41.0	36.9
VideoLLaMA2 [2024]	336^2	47.9	48.4
LLaVA-Next (StreamFormer)	224^2	**45.0**	**48.4**

- Shows strong 224×224 results attributed to enhanced temporal and spatial perception in the backbone.

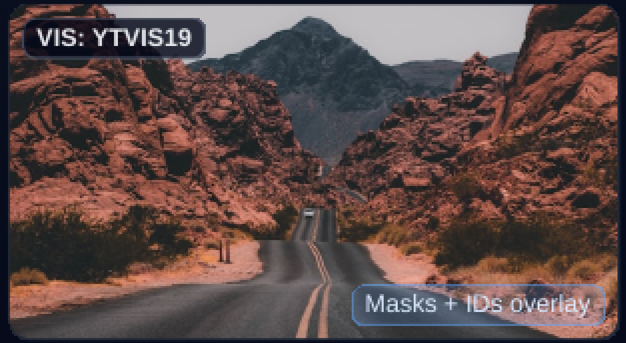
Backbones integrated into LLaVA-based pipelines [63, 124].

Table 3

ABLATIONS

Effect of Pre-training Tasks

- Any video-language pretraining improves over the SigLIP baseline.
- Global- and temporal-level tasks yield broad gains.
- Combining global, temporal, and spatial tasks provides the best overall performance.



VideoQA frames with answer

VideoQA

Question and predicted answer

Multitask pretraining improves outputs across tasks.

Table 4

ABLATIONS

Temporal Attention and Scalability

ACCURACY (W/O FLOW)

TA Type	mAP
No TA	61.7
Bi-directional	62.8
Causal	**62.5**

EFFICIENCY

Latency vs frames plot

KV-Cache

Causal TA scales with near-constant latency.

- Causal temporal attention attains near best accuracy with superior streaming efficiency.

Table 5 • Fig.3

EFFICIENCY

Data Efficiency vs WebVid

Dataset	OAD w/ Flow	OAD w/o Flow	VideoMME	MLVU
WebVid-1M	69.3	57.2	37.4	46.6
StreamFormer-0.1M	73.0	62.5	43.1	49.6
StreamFormer-1M	**73.9**	**68.4**	**45.0**	**48.4**

- High-quality annotated datasets enable strong performance with less data.

DOWNSTREAM

Task Heads and Integrations

- OAD: MAT with StreamFormer features (optionally concatenated with optical flow).
- Online VIS: CTVIS with ViT-Adapter; StreamFormer backbone frozen.
- VideoQA: LLaVA-Next with StreamFormer (224^2), identical pipeline as prior work.

DISCUSSION

Open Questions

- How can the backbone scale to hour-long or continuous streams while preserving low latency?
- What mechanisms enable robust multi-camera or multi-sensor fusion in streaming settings?
- How resilient is causal attention to distribution shift, occlusion, and drift over time?
- Can memory modules or KV-Cache compression improve long-horizon reasoning without accuracy loss?
- What are the trade-offs between open-set language alignment and fine-grained spatial precision?

REFERENCES

Key Resources

- SigLIP: Multimodal pretraining for vision-language alignment [119].
- Streaming tasks: OAD (THUMOS-14), YTVIS-2019, VideoQA (VideoMME, MLVU).
- Project: https://go2heart.github.io/streamformer
- Final published version on IEEE Xplore; OA via CVF.

CITATIONS USED IN TABLES

- MAT [96] (Online Action Detection baseline).
- LSTR [110] (Streaming temporal modeling for OAD).
- GateHUB [16] (Gated streaming OAD).
- CTVIS [116] (Online Video Instance Segmentation).
- LLaVA [63], LLaVA-Next-Video [124], VideoLLaMA2 [21] (VideoQA baselines).